

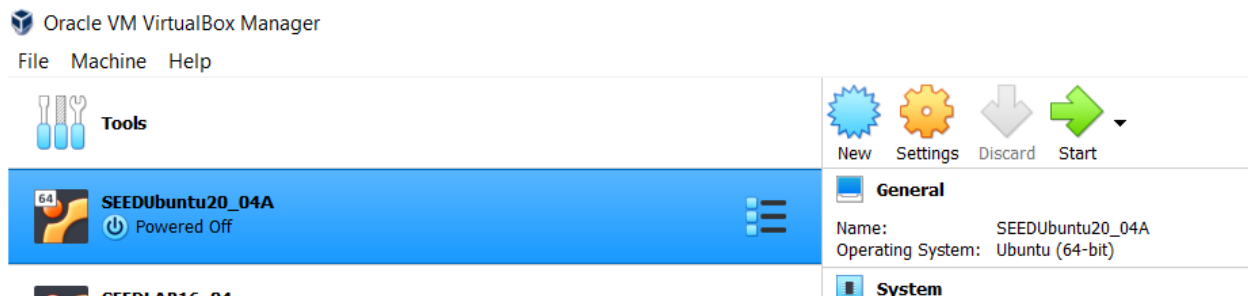
Lab 1

Due May 13

In this lab, you need to complete the following tasks. In your solution, whenever applicable, you need to provide screenshots to prove your work. **In this course**, your solution without screenshots should be a **complete** answer to the question and the screenshot only works as an **evidence**, unless the screenshot itself is the required answer. Screenshots should mark **elements** that are **essential** in your solution.

SEEDVM Setup on your computer: SEEDUbuntu 20.04 at <https://seedsecuritylabs.org/labsetup.html> (but do not use Ubuntu 16.04); for **Cloud setup** on AWS, follow my file.

1. Complete the SEEDLAB setup. You can set up on your computer or cloud (if your computer is not possible). Provide a screen shot as a proof of completing your task. Here is an example.



Warning: In this problem, the screen shot is certainly not a sound evidence for your task completion. You need to be honest. Successful completion of SEEDVM is essential for this course.

2. This question considers sniffing http packet.

- Start your Wireshark capture.
- On your browser, access website of your choice. I recommend <http://www.example.com> (as it is simple and it is easy to find out on the Wireshark packet list).
- Stop Wireshark

Answer the following questions. In your solution, the screen shot needs to mark the ip addresses required in the questions. If you can not find the answer, try: (1) clear your browser history and (2) clear your ubuntu DNS cache `$ sudo systemd-resolve --flush-caches`

a. In the packet list, the column **Protocol** contains many protocols. Specify at least 4 of them.

b. open one DNS packet. You should see link layer, network layer, transport layer and application layer, as discussed in class. Give each layer a screenshot.

c. Before your browser sends a request to the webserver, it first issue DNS query to find out the ip address of this website. Give a screenshot for this DNS query packet and the response packet. Note: in the query packet, you should be information such as **[Response In: 14]**. This 14 is the packet index of the response packet. In the response packet, click on **answers**. Then, ip address of the website will be there. What is the ip address of this website? You can apply filter **dns** in your Wireshark to restrict the packet list to DNS packets only.

d. The source IP address of the DNS query packet will be your IP address and the destination IP address of the DNS query packet will be the DNS server's IP address. What is the ip address of this DNS server? What is the ip address of your machine?